

CS 173 — Discrete Structures

Lectures 24–35: A Unified Treatment

Modular Arithmetic, Functions, Cardinality,
and Graph Theory

These notes consolidate Lectures 24 and 26–35 into a single coherent arc: from equivalence relations (modular congruence) through functions and cardinality to applications in graph theory. Each part builds directly on the previous one; reading in order is recommended.

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Part I

Modular Arithmetic

This part introduces the first non-trivial example of an equivalence relation we'll meet in the course: *congruence modulo n* . The algebraic structure $(\mathbb{Z}_n, +, \cdot)$ that emerges is a useful object in its own right, and it foreshadows a recurring theme — defining “equality up to” a structural relation.

1 Congruence modulo n

Definition 1.1 (Congruence modulo n). For $x, y \in \mathbb{Z}$ and $n \in \mathbb{N}^+$, x is *congruent to y modulo n* , written $x \equiv y \pmod{n}$, iff $n \mid (x - y)$.

Remark. Read $\equiv$ as a structured equality: rather than asking whether two integers are identical, we ask whether they leave the same remainder when divided by n .

Theorem 1.1 ($\equiv \pmod{n}$ is an equivalence relation). For all $n \in \mathbb{N}^+$:

- (i) **Reflexive:** $x \equiv x \pmod{n}$ since $n \mid 0$.
- (ii) **Symmetric:** $x \equiv y \Rightarrow y \equiv x$, since $n \mid (x - y) \Rightarrow n \mid -(x - y)$.
- (iii) **Transitive:** $x \equiv y$ and $y \equiv z$ imply $x \equiv z$, since $n \mid (x - y) + (y - z) = x - z$.

2 Compatibility with arithmetic

Theorem 2.1 (Compatibility). For all $n \in \mathbb{N}^+$ and all $x, y, z \in \mathbb{Z}$, if $x \equiv y \pmod{n}$ then

$$x + z \equiv y + z \pmod{n} \quad \text{and} \quad x \cdot z \equiv y \cdot z \pmod{n}.$$

Theorem 2.2 (Cancellation). **Additive cancellation** holds unconditionally:

$$x + z \equiv y + z \pmod{n} \Rightarrow x \equiv y \pmod{n}.$$

Multiplicative cancellation holds only when $\gcd(z, n) = 1$:

$$x \cdot z \equiv y \cdot z \pmod{n} \wedge \gcd(z, n) = 1 \Rightarrow x \equiv y \pmod{n}.$$

Caution (Multiplicative cancellation can fail). Without the coprimality hypothesis, multiplicative cancellation breaks. E.g. in $\mathbb{Z}_6$: $2 \cdot 2 = 4$ and $2 \cdot 5 = 10 \equiv 4$, so $2 \cdot 2 \equiv 2 \cdot 5 \pmod{6}$, but $2 \not\equiv 5 \pmod{6}$. The issue: $\gcd(2, 6) = 2 \neq 1$.

3 The ring $\mathbb{Z}_n$

Definition 3.1 (Integers modulo n). For $n \in \mathbb{N}^+$, define $\mathbb{Z}_n := \{0, 1, 2, \dots, n-1\}$, the set of *residues* modulo n . Define modular operations on $\mathbb{Z}_n$ by

$$a \oplus b := (a + b) \bmod n, \quad a \otimes b := (a \cdot b) \bmod n.$$

Example (Working in $\mathbb{Z}_6$). The additive inverses pair up:

$$1 + 5 \equiv 0, \quad 2 + 4 \equiv 0, \quad 3 + 3 \equiv 0 \pmod{6}.$$

Each pair $(a, 6 - a)$ sums to 0.

Theorem 3.1 (Algebraic properties of $\mathbb{Z}_n$). $(\mathbb{Z}_n, \oplus, \otimes)$ is a *commutative ring*: $\oplus$ and $\otimes$ are commutative and associative, $\otimes$ distributes over $\oplus$, 0 is the additive identity, 1 is the multiplicative identity, and every $a \in \mathbb{Z}_n$ has additive inverse $n - a$.

Caution (Multiplicative inverses are not guaranteed). $a \in \mathbb{Z}_n$ has a multiplicative inverse iff $\gcd(a, n) = 1$. When n is prime, every nonzero element is invertible and $\mathbb{Z}_n$ is a *field*.

Remark (Looking ahead). The ring properties of $\mathbb{Z}_n$ underpin the solution of linear congruences $ax \equiv b \pmod{n}$ via the extended Euclidean algorithm. The deeper takeaway, though, is the pattern: an equivalence relation on $\mathbb{Z}$ produces a quotient set $\mathbb{Z}_n$ that inherits a coherent algebraic structure. This pattern reappears throughout discrete mathematics.

Part II

Functions

We now turn to the language of functions. Three classes of functions — injections, surjections, and bijections — form the bedrock for everything that follows. In Part III we use them to compare the sizes of sets, finite or infinite.

4 Injections, surjections, bijections

Throughout this section let A, B be sets and $f: A \rightarrow B$ a function.

Definition 4.1 (Injection / one-to-one). f is *injective* iff distinct inputs produce distinct outputs:

$$\forall a_1, a_2 \in A, f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

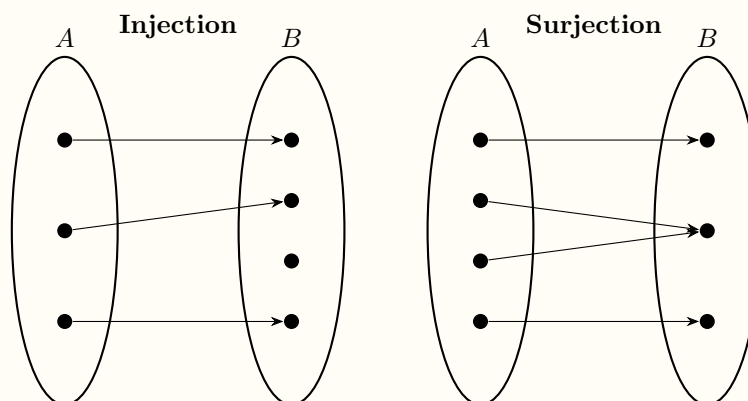
Equivalently (by contraposition), $a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$.

Definition 4.2 (Surjection / onto). f is *surjective* iff every element of B is hit:

$$\forall y \in B, \exists x \in A, f(x) = y.$$

Definition 4.3 (Bijection). f is a *bijection* iff it is both injective and surjective.

4.1 Diagrammatic intuition



Property	Injection	Surjection
Statement	$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$	$\forall y \in B, \exists x \in A, f(x) = y$
Constraint on	inputs (domain A)	outputs (codomain B)
Violation	two arrows into one dot in B	a dot in B with no arrow

5 Inverses, monomorphisms, epimorphisms

Definition 5.1 (Identity, left/right inverse). The *identity function* $\text{id}_A: A \rightarrow A$ is $\text{id}_A(a) = a$. A function $g: B \rightarrow A$ is a *left inverse* of $f: A \rightarrow B$ if $g \circ f = \text{id}_A$, and a *right inverse* if $f \circ g = \text{id}_B$.

Definition 5.2 (Mono / epi / iso). Let $f: A \rightarrow B$.

- f is a *monomorphism* (or *monic*) iff f has a left inverse.
- f is an *epimorphism* (or *epic*) iff f has a right inverse.
- f is an *isomorphism* iff f is both monic and epic.

Theorem 5.1 (Inverses from injections / surjections). Let $f: A \rightarrow B$.

- (I) If f is injective and $A \neq \emptyset$, there exists a *surjective* $g: B \rightarrow A$ with $g \circ f = \text{id}_A$.
- (II) If f is surjective, there exists an *injective* $g: B \rightarrow A$ with $f \circ g = \text{id}_B$.

Caution (Empty-domain corner case). Part (I) fails when $A = \emptyset$ and $B \neq \emptyset$: the empty function is vacuously injective, but no $g: B \rightarrow \emptyset$ exists.

Corollary 5.2 (Two-sided inverse of a bijection). If $f: A \rightarrow B$ is a bijection, there exists a *unique* bijection $g: B \rightarrow A$ satisfying both $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$. This g is denoted f^{-1} .

Remark. Uniqueness fails on one side alone — a non-surjective injection has many left inverses (any preimage choice on $B \setminus f(A)$). Two-sidedness is what pins g down.

5.1 Cancellation characterisation

The same three notions can be characterised *purely by composition*, without referring to elements.

Theorem 5.3 (Cancellation forms). Let $f: A \rightarrow B$.

- (1) f is injective $\iff \forall W, h_1, h_2: W \rightarrow A, f \circ h_1 = f \circ h_2 \Rightarrow h_1 = h_2$.
- (2) f is surjective $\iff \forall Z, k_1, k_2: B \rightarrow Z, k_1 \circ f = k_2 \circ f \Rightarrow k_1 = k_2$.
- (3) f is a bijection $\iff f$ has a two-sided inverse.

Remark (Why this matters). The “element-level” definitions look inside A and B ; the cancellation forms only mention compositions. This generalises to settings where objects don’t have “elements” — the entry point to category theory.

Part III

Cardinality

We use injections and surjections to compare set sizes — a definition that works equally well for finite and infinite sets.

6 Comparing cardinalities

Definition 6.1 (Cardinality comparisons). For sets A, B :

- $|A| \leq |B| \iff \exists f: A \rightarrow B$ injective.
- If $B \neq \emptyset$: $|A| \geq |B| \iff \exists f: A \rightarrow B$ surjective.
Empty target convention: $|A| \geq |\emptyset|$ for all A .
- $|A| = |B| \iff \exists f: A \rightarrow B$ a bijection $\iff |A| \leq |B|$ and $|A| \geq |B|$.

Remark (Why the asymmetry). An injection $A \hookrightarrow B$ “fits” all of A into B without collisions, so B must be at least as large. A surjection $A \twoheadrightarrow B$ “covers” all of B , so A must be at least as large. The empty target is a degenerate case: no surjection $A \rightarrow \emptyset$ exists when $A \neq \emptyset$, so we patch the definition by fiat.

7 Order-theoretic properties

Theorem 7.1 (Order properties of $\leq$). For all sets A, B, C :

- Reflexive**: $|A| = |A|$ (witnessed by id_A).
- Duality**: $|A| \leq |B| \iff |B| \geq |A|$.
- Transitive**: $|A| \leq |B|$ and $|B| \leq |C| \Rightarrow |A| \leq |C|$ (witnessed by composition).
- Antisymmetric (Cantor–Schröder–Bernstein)**: $|A| \leq |B|$ and $|A| \geq |B| \Rightarrow |A| = |B|$.
- Trichotomy** (Axiom 7, equivalent to AC): $|A| \leq |B|$ or $|A| \geq |B|$.

Remark (CSB is the deep part). For finite sets, antisymmetry follows from counting. For infinite sets, it requires *constructing* a bijection from two given injections — the content of CSB. Trichotomy is independent of ZF, equivalent to AC.

Definition 7.1 (Strict cardinality). $|A| < |B|$ iff there is an injection $A \rightarrow B$ but no surjection $A \rightarrow B$. Symmetrically, $|A| > |B|$ iff there is a surjection $A \rightarrow B$ but no injection $A \rightarrow B$.

Remark. Each strict definition pairs an existential witness with a universal failure. The universal clause cannot be skipped — a single non-surjective function (e.g. a constant) proves nothing on its own.

8 Finite sets

Definition 8.1 (Finite set, cardinality of a finite set). Identify $n \in \mathbb{N}$ with the set $\{0, 1, \dots, n-1\}$. A set X is *finite* iff $|X| = |n|$ for some $n \in \mathbb{N}$. The unique such n is then called $|X|$.

Remark (Why n is unique). Distinct naturals $m \neq n$ admit no bijection between them, so antisymmetry forces the matching n to be unique.

8.1 Counting laws

Theorem 8.1 (Counting laws on finite sets). If X, Y are finite then so are $X \cap Y$, $X \cup Y$, $X \setminus Y$, and $X \times Y$. Moreover:

- (i) *Subset subtraction*: If $Y \subseteq X$, then $|X \setminus Y| = |X| - |Y|$.
- (ii) *General subtraction*: $|X \setminus Y| = |X| - |X \cap Y|$.
- (iii) *Inclusion-exclusion (two-set)*: $|X \cup Y| = |X| + |Y| - |X \cap Y|$.
- (iv) *Union bound*: $|\bigcup_{i=1}^n A_i| \leq \sum_{i=1}^n |A_i|$, with equality iff the A_i are pairwise disjoint.
- (v) *Product*: $|X \times Y| = |X| \cdot |Y|$.

8.2 Single-element removal

The following lemma is the key induction step in many counting arguments; the proof is included because the technique — gap-closing relabelling — recurs throughout the course.

Lemma 8.2 (Removing one element). For any finite set X and $a \in X$, $|X \setminus \{a\}| = |X| - 1$.

Proof. Let $|X| = n$, fix a bijection $f: X \rightarrow n = \{0, \dots, n-1\}$. Define $g: X \setminus \{a\} \rightarrow n-1$ by

$$g(b) = \begin{cases} f(b) & \text{if } f(b) < f(a), \\ f(b) - 1 & \text{if } f(b) > f(a). \end{cases}$$

The two cases are mutually exclusive because $b \neq a$ and f is injective. We verify g is a bijection.

Injective. Suppose $g(b_1) = g(b_2)$. If $f(b_1) < f(a)$ and $f(b_2) > f(a)$, then $g(b_1) = f(b_1) < f(a)$ but $g(b_2) = f(b_2) - 1 \geq f(a)$, a contradiction. So $f(b_1)$ and $f(b_2)$ lie on the same side of $f(a)$, in which case $g(b_1) = g(b_2)$ forces $f(b_1) = f(b_2)$, hence $b_1 = b_2$ by injectivity of f .

Surjective. Let $y \in n - 1$. If $y < f(a)$, then since f is surjective there is $b \in X$ with $f(b) = y$; we have $b \neq a$ (else $y = f(a)$), so $g(b) = f(b) = y$. If $y \geq f(a)$, then $y + 1 \in n$ and there is $b \in X$ with $f(b) = y + 1 > f(a)$; again $b \neq a$, and $g(b) = f(b) - 1 = y$.

Hence g is a bijection, so $|X \setminus \{a\}| = n - 1$. $\square$

9 Pigeonhole principle

Theorem 9.1 (Pigeonhole). Let A, B be sets.

- (i) $|A| > |B| \Rightarrow$ no $f: A \rightarrow B$ is injective.
- (ii) $|A| < |B| \Rightarrow$ no $f: A \rightarrow B$ is surjective.

Theorem 9.2 (Quantitative pigeonhole). Let $|A| = n \in \mathbb{N}^+$, $|B| = k \in \mathbb{N}^+$, $f: A \rightarrow B$. Then some fibre is large:

$$\exists b \in B, \quad |\{a \in A \mid f(a) = b\}| \geq \left\lceil \frac{n}{k} \right\rceil.$$

Remark (Why the bound). If every fibre had size $< \lceil n/k \rceil$, the total $\sum_b |f^{-1}(\{b\})|$ would be at most $k(\lceil n/k \rceil - 1) < n$, contradicting that the fibres partition A and so sum to $|A| = n$.

Proposition 9.3 (Finite shortcut). For *finite* A, B :

- An injective non-surjection $A \rightarrow B$ witnesses $|A| < |B|$.
- A surjective non-injection $A \rightarrow B$ witnesses $|A| > |B|$.

Caution (Infinite sets break the shortcut). Take $f: \mathbb{N} \rightarrow \mathbb{N}$, $n \mapsto 2n$: injective, not surjective, yet $|\mathbb{N}| = |\mathbb{N}|$. For strict comparisons in the infinite case, the universal clause must be proved directly — the job of Cantor's diagonal.

10 Infinite sets

Definition 10.1 (Infinite, Dedekind infinite). A set X is *infinite* iff X is not finite, i.e. $\forall n \in \mathbb{N}$, $|X| \neq |n|$.

X is *Dedekind infinite* iff some *proper* subset $Y \subsetneq X$ satisfies $|Y| = |X|$.

Theorem 10.1 (Equivalence). X is infinite $\iff X$ is Dedekind infinite.

Remark (Status of the proof). The $(\Leftarrow)$ direction is straightforward. The $(\Rightarrow)$ direction requires the Axiom of Choice in full generality. The two constructions below witness the Dedekind side concretely for $\mathbb{N}$.

10.1 $\mathbb{N}$ is genuinely infinite

Theorem 10.2. $\mathbb{N}$ is infinite.

Proof. Suppose, for contradiction, that there is a bijection $f: n \rightarrow \mathbb{N}$ for some $n \in \mathbb{N}$. Let $S = \sum_{i=0}^{n-1} f(i) \in \mathbb{N}$. Each $f(i)$ is non-negative, so $f(i) \leq S < S+1$ for every i . By surjectivity of f , some $x \in n$ satisfies $f(x) = S+1$, contradicting $f(x) \leq S$. $\square$

10.2 Bijections between countably infinite sets

Theorem 10.3 (Same-size infinities). $|\mathbb{N}| = |\mathbb{N}^+| = |\mathbb{Z}| = |E| = |\mathbb{N} \times \mathbb{N}|$, where E is the set of even integers.

Sketch. Each equality is witnessed by an explicit bijection; combined with the identity injections in the obvious directions, Cantor–Schröder–Bernstein finishes the harder cases.

$|\mathbb{N}| = |\mathbb{N}^+|$: $n \mapsto n+1$.

$|\mathbb{Z}| = |\mathbb{N}|$: interleave non-negatives and negatives,

$$f(x) = \begin{cases} 2x & x \geq 0, \\ -2x-1 & x < 0. \end{cases}$$

First several values: $0 \mapsto 0$, $-1 \mapsto 1$, $1 \mapsto 2$, $-2 \mapsto 3$, $2 \mapsto 4$, $\dots$

$|\mathbb{N}| = |E|$: analogously, $f: \mathbb{N} \rightarrow E$ given by $f(n) = n$ if n even, $f(n) = -n-1$ if n odd.

$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$: the inclusion $n \mapsto (0, n)$ gives one injection. The encoding $(x, y) \mapsto 2^x \cdot 3^y$ gives the other — injective by unique prime factorisation, since 2 and 3 are distinct primes. CSB closes the argument. $\square$

Caution (Bad encodings). Tempting alternatives like $(x, y) \mapsto 2^x + 3^y$ or $(x, y) \mapsto x + y$ are *not* injective. Multiplicativity together with distinct prime bases is exactly what unique factorisation needs.

10.3 Hilbert’s hotel

The bijections above admit a vivid retelling. Imagine a hotel with rooms labelled $\mathbb{N}^+$, every room occupied by guest G_n .

Scenario	Reassignment	Underlying bijection
1 new guest	$G_n \rightarrow \text{room } n + 1$	$n \mapsto n + 1$
Bus of $\mathbb{N}^+$ guests	$G_n \rightarrow \text{room } 2n, H_k \rightarrow \text{room } 2k - 1$	$\mathbb{N}^+ \sqcup \mathbb{N}^+ \rightarrow \mathbb{N}^+$

Caution (Simultaneous, not sequential). Hilbert’s reassignment is a single global act, not a process. There is no “last” guest to move, so the bijection cannot be built one step at a time.

11 Cantor’s theorem and the hierarchy of infinities

So far every countably infinite set we’ve seen has $|\mathbb{N}|$. Cantor shows the power set genuinely grows the cardinality.

Theorem 11.1 (Cantor). $|X| < |\mathcal{P}(X)|$ for every set X .

Diagonal argument. The map $x \mapsto \{x\}$ is an injection $X \rightarrow \mathcal{P}(X)$, so $|X| \leq |\mathcal{P}(X)|$. For the strict inequality, we show no $\varphi: X \rightarrow \mathcal{P}(X)$ is surjective.

Define the *anti-diagonal set* $\Delta = \{x \in X \mid x \notin \varphi(x)\} \in \mathcal{P}(X)$. Suppose φ were surjective. Then $\varphi(\delta) = \Delta$ for some $\delta \in X$, and the biconditional

$$\delta \in \Delta \iff \delta \notin \varphi(\delta) \iff \delta \notin \Delta$$

is a contradiction. So φ is not surjective. □

Remark (Connections). The same self-referential trick produces *Russell’s paradox* (let X be “the set of all sets”, φ the identity). The contradiction in Cantor’s proof becomes a hard contradiction in naive set theory, which is why ZFC excludes a universal set.

The diagonal name comes from imagining φ as an infinite table: row x , column y records whether $y \in \varphi(x)$. Δ is built by walking down the diagonal and *flipping* each entry, so it disagrees with every row.

Remark (The hierarchy of infinities). Iterating $\mathcal{P}$ gives a strictly increasing chain of cardinalities:

$$|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < \dots$$

The first jump is the canonical leap from countable to uncountable: $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}|$.

Part IV

Applications

The cardinality machinery now pays off. In §12 we use pigeonhole to prove a classic combinatorial identity about parties; in §13 we set up graph theory and prove the handshaking lemma by induction.

12 The handshake theorem

Theorem 12.1 (Handshake theorem). Any room with at least two people contains two distinct people who have shaken the same number of hands.

Setup

Fix $n \geq 2$. Take $A = \{0, 1, \dots, n-1\}$ for the set of people ($|A| = n$), and a symmetric handshake relation $H \subseteq A \times A$. Define

$$f: A \rightarrow B, \quad f(p) = |\{a \in A \mid (p, a) \in H\}|,$$

where $B \subseteq D = \{0, 1, \dots, n-1\}$ is the set of *actual* handshake counts that occur. The proof reduces to showing $|B| < |A|$, then quoting pigeonhole.

The codomain bound

Lemma 12.2 (Codomain bound). $|B| \leq n-1$.

Proof. Either everyone has shaken ≥ 1 hands or someone has shaken 0.

Case 1: $0 \notin B$. Then $B \subseteq D \setminus \{0\}$, so $|B| \leq n-1$.

Case 2: *someone has shaken 0 hands.* Fix p with $f(p) = 0$. We show $n-1 \notin B$ by contradiction.

Suppose $f(q) = n-1$ for some q . Then q has shaken hands with *everyone else* — the set $\{a \in A \mid (q, a) \in H\} \subseteq A \setminus \{q\}$ has size $n-1 = |A \setminus \{q\}|$, so equality holds. In particular, $p \neq q$ (else $f(p) = f(q) = n-1 \neq 0$), so $p \in A \setminus \{q\}$ and hence $(q, p) \in H$. By symmetry $(p, q) \in H$, so $f(p) \geq 1$, contradicting $f(p) = 0$.

So $n-1 \notin B$, giving $|B| \leq n-1$. □

Remark. The two cases knock out different elements of D : 0 in Case 1, $n-1$ in Case 2. The combined fact is that $\{0, n-1\}$ cannot both lie in B — a universal hand-shaker and a hand-avoider can't coexist by symmetry of H .

Finishing with pigeonhole

Proof of the handshake theorem. $|B| < |A|$ by the lemma, so by pigeonhole f is not injective: there exist distinct $p_1, p_2 \in A$ with $f(p_1) = f(p_2)$. $\square$

Caution ($n \geq 2$ is needed). With $n = 1$ the statement is vacuously *false* (“two distinct people” don’t exist). The proof breaks too: Case 2 would need a single person to satisfy $f(p) = 0$ and have shaken everyone else’s hand, which is contradictory only because the universe “everyone else” is empty.

Remark (The template). This is a template, not a one-off result. Many “in any group of n objects, two must share a property” proofs follow the same recipe: build a counting function into a codomain that is provably one element smaller than the domain, then quote pigeonhole. The work is finding the structural reason a particular value can’t occur.

13 Graph theory

13.1 Graphs, neighbourhoods, degree

Definition 13.1 (Graph). A *graph* G is a pair $(V(G), E(G))$ where $V(G)$ is a finite set of *vertices* (or *nodes*) and $E(G)$ is a set of *edges* satisfying

$$E(G) \subseteq \{W \mid W \subseteq V(G) \wedge |W| = 2\}.$$

Remark (What the definition rules out). The constraint $|W| = 2$ forbids loops $\{v, v\} = \{v\}$. The set structure of $E(G)$ forbids parallel edges. Edges are *unordered* pairs, so the graph is undirected. $V(G)$ is finite, forcing $E(G) \subseteq \mathcal{P}(V(G))$ to be finite as well.

Caution. A graph is determined by the *pair* $(V(G), E(G))$, not by $E(G)$ alone. Isolated vertices (in $V(G)$ but in no edge) are invisible from the edge set.

Definition 13.2 (Neighbourhood, incident edges, degree). For $v \in V(G)$:

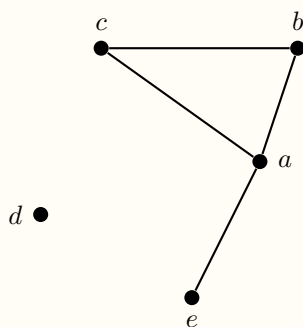
$$\begin{aligned} N_G(v) &:= \{w \in V(G) \mid \{v, w\} \in E(G)\} && \text{(neighbours of } v), \\ I_G(v) &:= \{e \in E(G) \mid v \in e\} && \text{(edges incident to } v), \\ \deg_G(v) &:= |I_G(v)| = |N_G(v)| && \text{(degree of } v). \end{aligned}$$

The two cardinalities agree because $w \mapsto \{v, w\}$ is a bijection $N_G(v) \rightarrow I_G(v)$.

Caution. $v \notin N_G(v)$ in our convention — this is the *open* neighbourhood. Some texts write $N_G[v] = N_G(v) \cup \{v\}$ for the *closed* version.

13.2 Worked example

Let $V(G) = \{a, b, c, d, e\}$, $E(G) = \{\{a, b\}, \{a, c\}, \{a, e\}, \{b, c\}\}$. The vertex d is *isolated*.



v	$N_G(v)$	$I_G(v)$	$\deg_G(v)$
a	$\{b, c, e\}$	$\{\{a, b\}, \{a, c\}, \{a, e\}\}$	3
b	$\{a, c\}$	$\{\{a, b\}, \{b, c\}\}$	2
c	$\{a, b\}$	$\{\{a, c\}, \{b, c\}\}$	2
d	$\emptyset$	$\emptyset$	0
e	$\{a\}$	$\{\{a, e\}\}$	1

Sum of degrees: $3 + 2 + 2 + 0 + 1 = 8 = 2 \cdot 4 = 2|E(G)|$. This is no coincidence.

13.3 The handshaking lemma

Definition 13.3 (Subgraph). $H \leq G$ iff $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.

Theorem 13.1 (Handshaking lemma). For every graph G ,

$$2 \cdot |E(G)| = \sum_{v \in V(G)} \deg_G(v).$$

Proof by induction on $|V(G)|$. Base ($n = 0$). $V(G) = \emptyset$ forces $E(G) = \emptyset$, and both sides equal 0.

Step. Assume the result for $|V(G)| = k$ and let $|V(G)| = k + 1$. Pick any $x \in V(G)$ and form H by deleting x and all edges incident to it:

$$V(H) = V(G) \setminus \{x\}, \quad E(H) = E(G) \setminus I_G(x) = \{e \in E(G) \mid x \notin e\}.$$

H is a graph with $|V(H)| = k$, so by the inductive hypothesis $2|E(H)| = \sum_{v \in V(H)} \deg_H(v)$. Two facts about how degrees change under this deletion:

(a) *Non-neighbours of x keep their degree.* For $v \in V(G) \setminus (N_G(x) \cup \{x\})$, every edge of G containing v also lies in H , so $\deg_H(v) = \deg_G(v)$.

(b) *Neighbours of x lose exactly one edge.* For $v \in N_G(x)$, the unique edge $\{v, x\}$ is removed, so $\deg_H(v) = \deg_G(v) - 1$.

Partitioning $V(G) = N_G(x) \sqcup (V(G) \setminus (N_G(x) \cup \{x\})) \sqcup \{x\}$:

$$\begin{aligned} \sum_{v \in V(G)} \deg_G(v) &= \sum_{v \in N_G(x)} (\deg_H(v) + 1) + \sum_{v \notin N_G(x) \cup \{x\}} \deg_H(v) + \deg_G(x) \\ &= |N_G(x)| + \sum_{v \in V(H)} \deg_H(v) + \deg_G(x) \\ &= \deg_G(x) + 2|E(H)| + \deg_G(x) \quad (\text{IH; } |N_G(x)| = \deg_G(x)) \\ &= 2(|I_G(x)| + |E(H)|) = 2|E(G)|, \end{aligned}$$

since $E(G)$ partitions into $I_G(x)$ and $E(H)$. $\square$

Corollary 13.2 (Parity). The number of odd-degree vertices in any graph is even.

Proof. $\sum_v \deg_G(v) = 2|E(G)|$ is even. Splitting the sum by parity, the contribution from even-degree vertices is even, so the contribution from odd-degree vertices is even too — and a sum of odd numbers is even iff there are an even number of summands. $\square$

Example. For the five-vertex graph above, the odd-degree vertices are a (degree 3) and e (degree 1): two of them, which is even. $\checkmark$

Remark (Two “handshakes”). The *handshake theorem* (§12) and the *handshaking lemma* (this section) are different results sharing a metaphor. Both ultimately depend on the same fact: each edge has two endpoints, so any quantity counted “per endpoint” double-counts edges.

Epilogue: the through-line

The four parts of this document trace a single arc. Modular arithmetic gave us our first equivalence relation and the lesson that “equality up to” a relation produces structured quotient sets. Functions formalised the language of *maps between sets*. Cardinality used those maps — specifically injections and surjections — to compare set sizes, extending counting to the infinite. The applications then deployed all this machinery: pigeonhole and graph theory both reduce to claims about how a function’s domain and codomain compare.

Remark (Looking ahead). The same toolkit drives a great deal of the rest of CS 173 and beyond: relations and partial orders generalise the equivalence patterns of Part I, solving linear congruences applies the cancellation theorems of Part II, and graph theory builds toward connectivity, trees, and graph isomorphism — which are themselves “cardinality of structures” questions in disguise.